[INCH−POUND]
A−A−59590/13E
15 August 2025
SUPERSEDING
A−A−59590/13D
13 July 2020

COMMERCIAL ITEM DESCRIPTION
SPECIFICATION SHEET

HOLDER, ELECTRICAL CARD, WEDGE RETAINERS, 3 PIECE, SCREW ACTUATED DRIVE, .375 X .375 INCH BODY SIZE, WEDGE MOUNTING BODY

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

The complete requirements for procuring electrical card holders described herein shall consist of this document and the latest issue in effect of A−A−59590.

CLASSIFICATION/PART IDENTIFICATION NUMBER (PIN). This commercial item description (CID) specification sheet uses a classification system which is included in the Part Identification Number (PIN) as shown in the following example (see notes herein).

AA59590	/13	B, C, H, R	*	* *	W, D, E
CID number	Applicable CID specification sheet	Finish	Mounting	Length (see table I)	Configuration (see table II)

Example: AA59590/13HV28W is the PIN for a hard black anodize finished, 2.8 inch (71.1 mm) long card holder. The card holder also features two tapped mounting holes for use with 2−56 UNC 2B fasteners and a lockwasher and flat washer under the screw head for added resistance to loosening.

SALIENT CHARACTERISTICS.

Performance. Card holders shall hold the circuit card assembly firmly in its installed position. When the card holder is installed properly, it is capable of withstanding 60g/6ms of shock, 25 G−rms of vibration, and provides from 2 to 4 degrees C/W/inch thermal resistance transfer between the circuit card assembly and the heat sink surfaces.

Material. Unless otherwise specified herein, the card holder materials shall be as specified in A−A−59590.

Interface and physical dimensions. The card holders supplied to this CID specification sheet shall be as specified on figures 1, 2, and 3, in table I, and A−A−59590.

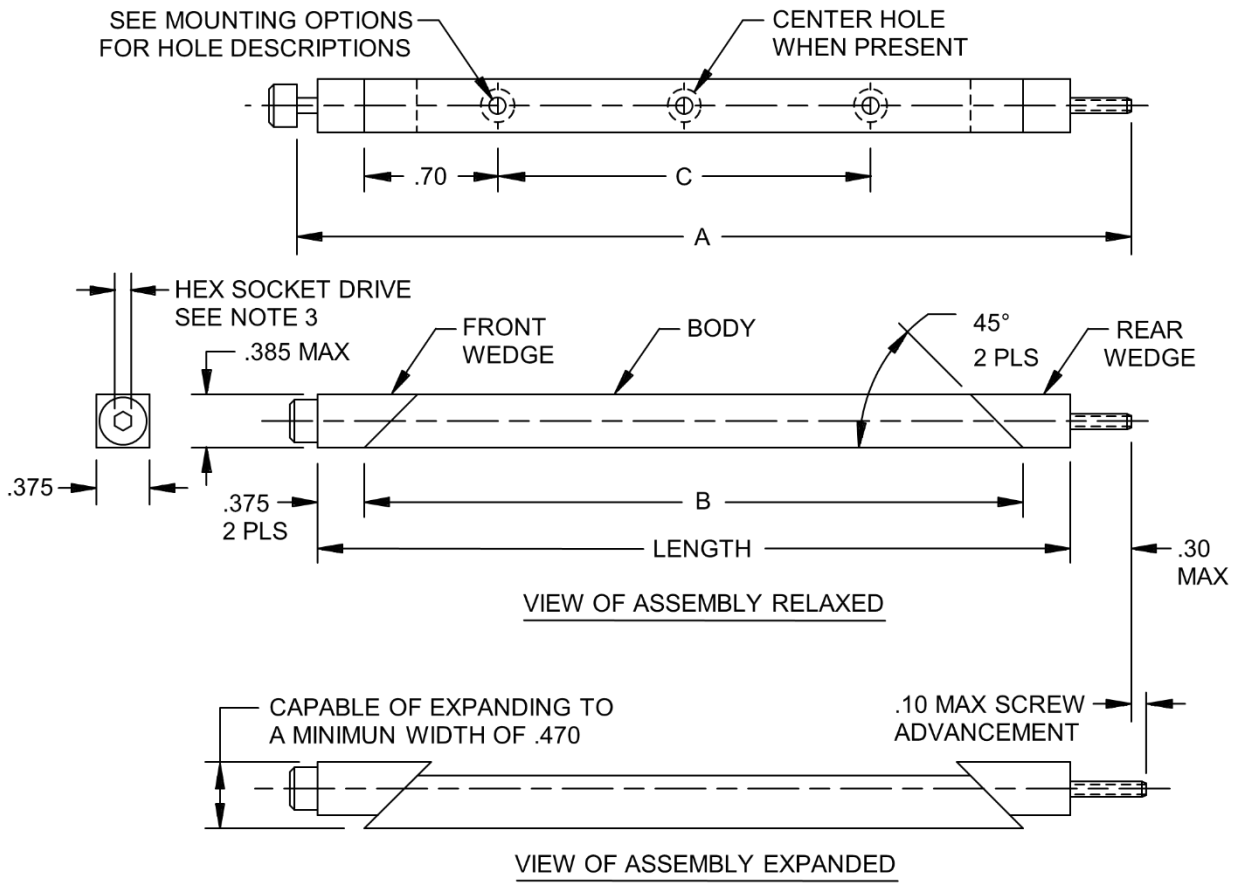
Actuating screw hex drive socket. The across flats dimension for the hex drive socket shall be as follows: .140 inch (3.56 mm) for mounting options "J", "N", "T", "R", "V", and "S" and 3.0 mm (0.118 inch) for mounting options "G" and "M".

Nominal installation torque. When card holders are used in cold plate applications, the nominal installation torque of each card holder shall be as follows: 16 to 20 inch-pounds (1.8 to 2.3 N-m) for assemblies using no options or configuration "W" and 17 to 21 inch-pounds (1.9 to 2.5 N-m) for assemblies of configurations "E" or "D".

Cold plate slot width dimensions (when applicable). The recommended cold plate slot width to accommodate the circuit card assembly with attached card holder is .425 inch (10.80 mm) plus the thickness of the printed board of the circuit card assembly (see A−A−59590).

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Inches	mm	Inches	mm	Inches	mm
.10	2.5	.375	9.5	.470	11.9
.20	5.1	.385	9.8	.70	17.8

NOTES:

1. Dimensions are in inches. Millimeters are given for general information only.
2. Unless otherwise specified, tolerances are ±.02 inch (0.51 mm) for two place decimals and ±.010 inch (0.25 mm) for three place decimals.
3. The across flats dimension for hex drive socket shall be .140 inch (3.56 mm) for mounting options "J", "N", "R", "T", "S" and "V". The across flats dimension for hex drive socket shall be a metric dimension of 3.0 mm (0.118 inch) across flats for mounting option "G" and "M".

FIGURE 1. Relaxed and expanded dimensions.

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Finish. The finish designator shall be as specified in A−A−59590. The finishes available for this CID specification sheet are as follows: "B" (black anodize), "C" (gold chemical film), "H" (hard black anodize), or "R" (clear chemical film).

Mounting. The mounting designators shall be as specified in A−A−59590. The mounting options available for this CID specification sheet are as follows: "J" (rivet mounting holes), "N" (no mounting holes), "T" or "R" (tapped 0−80 UNF 2B holes), "V" or "S" (tapped 2−56 UNC 2B holes), or "G" or "M" (tapped M2.5 x.45 metric holes). Card holders using mounting option "J" may be shipped unassembled. See figure 1 for mounting hole spacing requirements.

Mounting hole locations (when required). When mounting holes are specified in the PIN, two mounting holes are required on card holders less than 5.30 inches (134.6 mm) in length. Three mounting holes are required on card holders 5.30 inches (134.6 mm) or greater in length. The third mounting hole, when present, shall be centered on the mounting body. See figure 1 and table I for mounting hole locations and spacing requirements.

Rivet mounting holes. The holes used for rivet mounting shall be .136 inch (3.45 mm) diameter through holes, countersunk 100 degrees by .190 to .200 inch (4.83 to 5.08 mm) diameter with an access/clearance counterbore hole of .190 to .200 inch (4.83 to 5.08 mm) diameter by .300 inch (7.62 mm) deep.

Rivets. This card holder uses rivet style B as specified in A−A−59590 when rivet mounting is used.

Length, expanded, and relaxed dimensions. The length, expanded, and relaxed dimensions shall be as specified on figure 1. The length designator shall be as specified in A−A−59590 and the available lengths for this specification sheet are listed in table I.

TABLE I. Assembly dimensions (see figure 1). 1/

PIN length designator	Dimension "LENGTH" ±.02 (0.5)	Dimension "A" (maximum)	Dimension "B" ±.03 (0.8)	Dimension "C" ±.005 (0.13)
28	2.8 (71.1)	3.10 (78.7)	2.05 (52.1)	.65 (16.5)
33	3.3 (83.8)	3.60 (91.4)	2.55 (64.8)	1.15 (29.2)
38	3.8 (96.5)	4.10 (104.1)	3.05 (77.5)	1.65 (41.9)
43	4.3 (109.2)	4.60 (116.8)	3.55 (90.2)	2.15 (54.6)
48	4.8 (121.9)	5.10 (129.5)	4.05 (102.9)	2.65 (67.3)
53	5.3 (134.6)	5.60 (142.2)	4.55 (115.6)	3.15 (80.0)
58	5.8 (147.3)	6.10 (154.9)	5.05 (128.3)	3.65 (92.7)

1/ Dimensions are in inches. Millimeters, in parenthesis, are given for information only.

Configuration. The configuration of a card holder shall be as specified in table II. The details of a particular configuration consist of those on figure 1, and may include those on figures 2 and 3. Card holders not requiring the options described by table II shall leave the configuration position in the PIN blank.

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TABLE II. Configurations.

Configuration	Applicable figures	Hardware options
	1	No added options
W	1 and 2	Lockwasher and flat washer
E	1 and 3	Screw self-locking element
D	1, 2, and 3	Lockwasher, flat washer and screw self-locking element

Lockwasher and flat washer (see figure 2). A lockwasher and flat washer located under the screw head will provide for additional resistance to loosening of the card holder assembly from shock and vibration. Card holders requiring a lockwasher and flat washer shall include a suffix "W" in the PIN (see classification and notes).

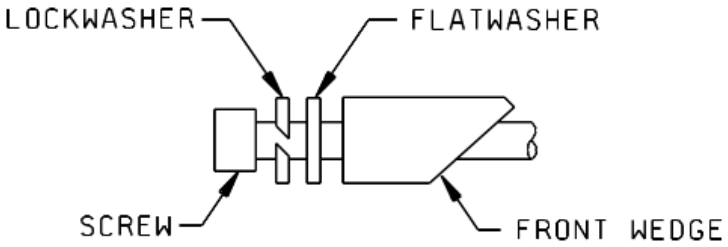


FIGURE 2. Lockwasher and flat washer details.

Screw self-locking element (see figure 3). The use of a screw self-locking element will provide prevailing torque for resistance to loosening from shock and vibration. Card holders requiring a screw self-locking element shall include a suffix "E" in the PIN (see classification and notes).

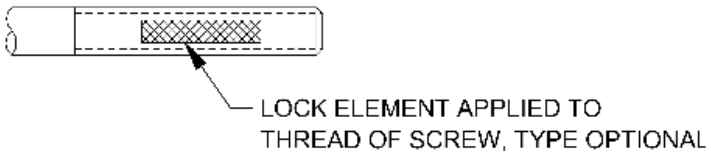


FIGURE 3. Screw self-locking element details.

Lockwasher, flat washer, and screw self-locking element. Card holders requiring a lockwasher, flat washer, and screw self-locking element configuration shall include a suffix "D" in the PIN (see classification and notes).

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NOTES.

PIN. The PIN should be used for Government purposes to buy commercial products to this CID specification sheet. See the classification section for PIN format example.

Source of documents.

Commercial Item Description

A−A−59590 − Holder, Electrical Card, Wedge Retainers, 3 Piece, for Cold Plate Applications, General Requirements For.

(Copies of these documents are available online at https://quicksearch.dla.mil.)

Ordering data. Ordering data shall be as specified in A−A−59590.

Commercial products. As part of the market analysis and research effort, this CID specification sheet was coordinated with the following manufacturers of commercial products. At the time of CID specification sheet preparation and coordination, these manufacturers were known to have commercial products that would meet the requirements of this CID specification sheet. (NOTE: This information should not be considered as a list of approved manufacturers or be used to restrict procurement to only the manufacturers shown.)

Manufacturer CAGE	Manufacturer name and address	Manufacturer contact information
5BG68	American Circuit Card Retainers, Inc. 2310 E. Orangethorpe Avenue Anaheim, CA 92806−1231	Telephone: (714) 738−6194 Facsimile: (714) 446−0119 E-mail: sales@accrmfg.com URL: https://www.accrmfg.com
61081	Schroff, Inc. 7328 Trade Street San Diego, CA 92121−3410	Telephone: (858) 740−2400 Facsimile: (858) 740−2430 E-mail: askschroff@nVent.com URL: https://schroff.nvent.com

Part number supersession data. This CID specification sheet PINs supersedes the following manufacturer's part numbers as shown in table III. The CID PINs listed in table III are only for length designator "28". See table IV for CID PIN construction using other available lengths for this specification sheet. This information is being provided to assist in reducing proliferation in the Government inventory system.

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TABLE III. Commercial part number supersession data.

PIN designator AA59590/13	Vendor similar designator or type part number 1/		
	CAGE 5BG68	CAGE 61081	
BJ28	3370BA-2.80H	A250-2.80H	
BN28	3370BA-2.80	A250-2.80	
BT28	3370BA-2.80T0	A250-2.80T0	
BV28	3370BA-2.80T2	A250-2.80T2	
BG28	3370MBA-2.80TM2.5	MA250-2.80TM2.5	
BJ28E	3370BA-2.80HL	A250-2.80HL	
BN28E	3370BA-2.80L	A250-2.80L	
BT28E	3370BA-2.80T0L	A250-2.80T0L	
BV28E	3370BA-2.80T2L	A250-2.80T2L	
BG28E	3370MBA-2.80TM2.5L	MA250-2.80TM2.5L	
BJ28W	3370WBA-2.80H	WA250-2.80H	
BN28W	3370WBA-2.80	WA250-2.80	
BT28W	3370WBA-2.80T0	WA250-2.80T0	
BV28W	3370WBA-2.80T2	WA250-2.80T2	
BG28W	3370MWBA-2.80TM2.5	MWA250-2.80TM2.5	
BJ28D	3370WBA-2.80HL	WA250-2.80HL	
BN28D	3370WBA-2.80L	WA250-2.80L	
BT28D	3370WBA-2.80T0L	WA250-2.80T0L	
BV28D	3370WBA-2.80T2L	WA250-2.80T2L	
BG28D	3370MWBA-2.80TM2.5L	MWA250-2.80TM2.5L	
CJ28	3370CG-2.80H	250-2.80H	
CN28	3370CG-2.80	250-2.80	
CT28	3370CG-2.80T0	250-2.80T0	
CV28	3370CG-2.80T2	250-2.80T2	
CG28	3370MCG-2.80TM2.5	M250-2.80TM2.5	
CJ28E	3370CG-2.80HL	250-2.80HL	
CN28E	3370CG-2.80L	250-2.80L	
CT28E	3370CG-2.80T0L	250-2.80T0L	
CV28E	3370CG-2.80T2L	250-2.80T2L	
CG28E	3370MCG-2.80TM2.5L	M250-2.80TM2.5L	

See footnotes at end of table.

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TABLE III. Commercial part number supersession data - Continued.

PIN designator AA59590/13	Vendor similar designator or type part number 1/		
	CAGE 5BG68	CAGE 61081	
CJ28W	3370WCG-2.80H	W250-2.80H	
CN28W	3370WCG-2.80	W250-2.80	
CT28W	3370WCG-2.80T0	W250-2.80T0	
CV28W	3370WCG-2.80T2	W250-2.80T2	
CG28W	3370MWCG-2.80TM2.5	MW250-2.80TM2.5	
CJ28D	3370WCG-2.80HL	W250-2.80HL	
CN28D	3370WCG-2.80L	W250-2.80L	
CT28D	3370WCG-2.80T0L	W250-2.80T0L	
CV28D	3370WCG-2.80T2L	W250-2.80T2L	
CG28D	3370MWCG-2.80TM2.5L	MW250-2.80TM2.5L	
HJ28	3370BH-2.80H	HA250-2.80H	
HN28	3370BH-2.80	HA250-2.80	
HT28	3370BH-2.80T0	HA250-2.80T0	
HV28	3370BH-2.80T2	HA250-2.80T2	
HG28	3370MBH-2.80TM2.5	MHA250-2.80TM2.5	
HJ28E	3370BH-2.80HL	HA250-2.80HL	
HN28E	3370BH-2.80L	HA250-2.80L	
HT28E	3370BH-2.80T0L	HA250-2.80T0L	
HV28E	3370BH-2.80T2L	HA250-2.80T2L	
HG28E	3370MBH-2.80TM2.5L	MHA250-2.80TM2.5L	
HJ28W	3370WBH-2.80H	WHA250-2.80H	
HN28W	3370WBH-2.80	WHA250-2.80	
HT28W	3370WBH-2.80T0	WHA250-2.80T0	
HV28W	3370WBH-2.80T2	WHA250-2.80T2	
HG28W	3370MWBH-2.80TM2.5	MWHA250-2.80TM2.5	
HJ28D	3370WBH-2.80HL	WHA250-2.80HL	
HN28D	3370WBH-2.80L	WHA250-2.80L	
HT28D	3370WBH-2.80T0L	WHA250-2.80T0L	
HV28D	3370WBH-2.80T2L	WHA250-2.80T2L	
HG28D	3370MWBH-2.80TM2.5L	MWHA250-2.80TM2.5L	

See footnotes at end of table.

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TABLE III. Commercial part number supersession data - Continued.

PIN designator AA59590/13	Vendor similar designator or type part number 1/		
	CAGE 5BG68	CAGE 61081	
RJ28	3370CC-2.80H	R250-2.80H	
RN28	3370CC-2.80	R250-2.80	
RT28	3370CC-2.80T0	R250-2.80T0	
RV28	3370CC-2.80T2	R250-2.80T2	
RG28	3370MCC-2.80TM2.5	MR250-2.80TM2.5	
RJ28E	3370CC-2.80HL	R250-2.80HL	
RN28E	3370CC-2.80L	R250-2.80L	
RT28E	3370CC-2.80T0L	R250-2.80T0L	
RV28E	3370CC-2.80T2L	R250-2.80T2L	
RG28E	3370MCC-2.80TM2.5L	MR250-2.80TM2.5L	
RJ28W	3370WCC-2.80H	WR250-2.80H	
RN28W	3370WCC-2.80	WR250-2.80	
RT28W	3370WCC-2.80T0	WR250-2.80T0	
RV28W	3370WCC-2.80T2	WR250-2.80T2	
RG28W	3370MWCC-2.80TM2.5	MWR250-2.80TM2.5	
RJ28D	3370WCC-2.80HL	WR250-2.80HL	
RN28D	3370WCC-2.80L	WR250-2.80L	
RT28D	3370WCC-2.80T0L	WR250-2.80T0L	
RV28D	3370WCC-2.80T2L	WR250-2.80T2L	
RG28D	3370MWCC-2.80TM2.5L	MWR250-2.80TM2.5L	

1/ The manufacturer's part number shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A-A-59590.

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PIN length examples. The CID PINs listed in table IV are for all available standard card holder lengths for this specification sheet. However, only one specific finish, mounting and configuration are listed (see PIN example for a break-down of the codes).

TABLE IV. Example of PIN with available length designators.

PIN designator AA59590/13	Vendor similar designator or type part number 1/ 2/		
	CAGE 5BG68	CAGE 61081	
BT28	3370BA−2.80T0	A250−2.80T0	
BT33	3370BA−3.30T0	A250−3.30T0	
BT38	3370BA−3.80T0	A250−3.80T0	
BT43	3370BA−4.30T0	A250−4.30T0	
BT48	3370BA−4.80T0	A250−4.80T0	
BR53	3370BA−5.30ET0	A250−5.30ET0	
BR58	3370BA−5.80ET0	A250−5.80ET0	

1/ The manufacturer's part number shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A−A−59590.

2/ Other lengths are available on request.

MILITARY INTERESTS:

Custodians:
Army − CR
Navy − EC
Air Force − 85
DLA − CC

CIVIL AGENCY COORDINATING ACTIVITY:

GSA − FAS

Preparing Activity:
DLA − CC

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